DATA CONSULTANCY REPORT

Customer Churn Prediction & Retention Strategy

Identifying at-risk customers and recovering $63,024 in preventable revenue loss through targeted intervention

Prepared by **Inez Agbenu | LinkedIn | GitHub**

Prepared for **TelcoX Marketing & Retention Team**

Date **June 2026**

Classification **Portfolio Project**

https://github.com/marzafiee/churn-risk-analyzer

Table of Contents

Executive Summary

Customer churn is a silent revenue drain. For TelcoX, analysis of 7,043 customer records reveals that 26.5% of customers churned and that the overwhelming majority left without any prior intervention. At an average monthly charge of $64.76, each lost customer represents approximately $777 in annual revenue. Across the full customer base, unmanaged churn represents millions in preventable annual loss.

This report presents the findings of a predictive churn analysis conducted on TelcoX customer data. Using machine learning, we identified which customers are most likely to leave, why they leave, and what a targeted retention campaign would cost and recover.

Headline finding: A tiered retention campaign targeting 601 high-risk customers is projected to generate **$63,024 in net retained revenue** at an overall **ROI of 699%** against a total campaign cost of $9,015.

High-risk customers deliver over 1,000% ROI and should be contacted immediately. The three primary drivers of churn are internet service type, contract length, and customer tenure, all actionable through existing retention levers.

1. Business Context & Problem Statement

1.1 Why churn matters

In subscription and service businesses, customer acquisition costs significantly more than customer retention. Industry research consistently finds that acquiring a new customer costs five to seven times more than retaining an existing one. For a telecom provider like TelcoX, this asymmetry makes churn not just a customer satisfaction issue but a core financial risk.

Churn erodes revenue in two ways. The direct loss is the monthly charge that disappears when a customer cancels. The indirect loss is the acquisition cost required to replace that customer — marketing spend, promotional offers, and onboarding resources that must be deployed to grow the customer base back to its previous size. A business that loses 26.5% of its customers annually is, in effect, running to stand still.

1.2 The problem this analysis solves

Most retention efforts fail not because the interventions are wrong but because they are applied too broadly or too late. Sending a discount offer to every customer is expensive and trains loyal customers to expect discounts. Waiting until a customer calls to cancel means the decision is already made.

The goal of this analysis is to answer three questions:

- Which customers are most likely to churn in the next 30–60 days?
- What is driving their likelihood to leave?
- What is the most cost-effective way to intervene?

1.3 Scope

This analysis covers 7,043 TelcoX customers across all service types, contract lengths, and demographics. The dataset captures customer behaviour, service usage, billing information, and churn outcome. The analytical approach is predictive, using historical patterns to score current customers by churn risk, rather than descriptive.

2. Data Overview

2.1 Dataset

The analysis was conducted on the IBM Telco Customer Churn dataset, a publicly available dataset representative of a mid-sized US telecom provider. It contains 7,043 customer records across 21 variables.

Property	Detail
Total customers	7,043
Churned customers	1,869 (26.5%)
Retained customers	5,174 (73.5%)
Average monthly charge	$64.76
Average tenure	32 months
Variables	21 original, 30 after encoding

2.2 Variables

The dataset captures four categories of customer information:

1. **Demographics**: gender, senior citizen status, partner, dependents
2. **Services subscribed:** phone service, multiple lines, internet service type, online security, online backup, device protection, tech support, streaming TV, streaming movies
3. **Account information**: tenure, contract type, paperless billing, payment method, monthly charges, total charges
4. **Outcome**: whether the customer churned

2.3 Data quality

The dataset was largely clean with one exception: the TotalCharges column loaded as a string type due to 11 blank entries representing new customers with no charge history. These were converted to numeric and null values filled with zero. No other missing values were present. The customerID column was dropped as it carries no predictive value.

2.4 Class imbalance

The dataset is imbalanced as 73.5% of customers did not churn, versus 26.5% who did. This is typical of churn datasets and has direct implications for model selection and evaluation. A naive model that predicts "will not churn" for every customer would achieve 73.5% accuracy while being completely useless. All modelling decisions in this analysis account for this imbalance explicitly.

3. Key Findings

3.1 Contract type is the strongest predictor of loyalty

Customers on month-to-month contracts churn at dramatically higher rates than those on annual or two-year contracts. The data shows a clear and consistent relationship: the longer the contract commitment, the lower the churn rate. Two-year contract holders are the most loyal segment by a significant margin.

This finding has an important strategic implication. Contract type is not just a predictor of churn but it is a retention lever. Converting a month-to-month customer to a one-year contract meaningfully reduces their probability of leaving. Retention campaigns that offer incentives for contract upgrades address the root cause rather than treating the symptom.

3.2 Fiber optic customers are the highest-risk premium segment

Customers subscribed to fiber optic internet service churn more than any other service group. This is the single largest driver of churn identified by the predictive model so it is more influential than any demographic or billing variable.

The pattern is counter-intuitive at first glance. Fiber optic is a premium service and these customers pay higher monthly charges. The likely explanation is a gap between expectation and experience so customers paying premium prices for premium service have higher expectations, and when those expectations are not met, they have both the motivation and the means to switch. This segment deserves targeted service quality investigation alongside retention outreach.

3.3 New, high-paying customers are the most urgent priority

Tenure, which is how long a customer has been with TelcoX, is a strong predictor of loyalty. Customers who have stayed for several years are significantly less likely to churn than those in their first year. The critical window is the first 12 months.

The 10 highest-risk customers identified by this analysis all have tenures of 7 months or less, with monthly charges between $70 and $106. These are new customers paying above-average prices who have not yet built the switching inertia that longer-tenured customers develop. Early intervention (proactive outreach in the first 3–6 months) is likely more effective and less costly than reactive retention after the decision to leave has already formed.

3.4 Payment method and additional services signal risk

Customers paying by electronic check churn at higher rates than those using automatic payment methods. Automatic payment customers have a passive retention mechanism built in so the friction of cancelling is higher. Electronic check customers require active monthly engagement with their bill, creating regular opportunities to reconsider.

Customers without online security or tech support add-ons also show higher churn rates. These services likely function as stickiness features, i.e. customers who have invested in additional services have more to lose by switching and more of their digital life tied to the provider.

3.5 Senior citizens and customers without dependents show elevated risk

Senior citizen customers and those without partners or dependents show slightly higher churn rates. These segments may have different price sensitivity, different support needs, or different decision-making processes. While these variables are less actionable than contract type or internet service, they are useful for segmentation — a retention offer for a senior citizen on a fixed income may need to be structured differently than one for a young professional.

4. Methodology

4.1 Analytical approach

This analysis follows a **supervised machine learning approach to binary classification.** The objective is to predict, for each customer, the probability that they will churn. This probability score becomes the basis for prioritising retention outreach. The analytical pipeline proceeded in **four** stages: data exploration, data cleaning and preparation, predictive modelling, and cost-benefit analysis. Each stage informed the next.

4.2 Why recall was chosen as the primary metric

Model performance in classification problems can be measured in multiple ways. For this business problem, **recall** (the proportion of actual churners that the model successfully identifies) is the priority metric.

The reasoning is straightforward. There are two types of error a churn model can make:

- **False negative:** A customer who churns that the model missed. Cost: full monthly revenue lost (~$65/month).
- **False positive:** A loyal customer flagged as a churn risk. Cost: an unnecessary retention offer (~$15).

<u>Missing a churner is more than four times more expensive than a false alarm.</u> This asymmetry means a model optimised purely for accuracy is the wrong tool so we need a model that catches as many real churners as possible, even at the cost of some additional false alarms.

4.3 Models evaluated

Three classification models were trained and compared:

➔ **Logistic Regression** was used as the interpretable baseline. Despite being the simplest model, **it performs well** when the relationship between features and the target is approximately linear which is partly the case here given the strong, clean signal from contract type and internet service.

➔ **Random Forest** was included as an ensemble method that builds multiple decision trees and aggregates their predictions. It can capture non-linear relationships and interactions between features. **In this analysis it underperformed**, likely due to over-reliance on three correlated financial variables: TotalCharges, MonthlyCharges, and tenure, which are not independent signals.

➔ **XGBoost** (Extreme Gradient Boosting) is a sequential tree-based method where each new tree corrects the errors of the previous one. It is widely used in industry for tabular classification problems and showed the **strongest overall performance after threshold tuning.**

4.4 Threshold tuning

Every classification model outputs a probability score, not just a yes/no prediction. By default, customers are flagged as churn risks when their probability exceeds 50%. Lowering this threshold flags more customers, thus, catching more real churners at the cost of more false alarms.

Given the cost asymmetry established in Section 4.2, we evaluated all three models across thresholds from 0.10 to 0.55. The optimal threshold was identified as the point where recall exceeded 0.75 without precision falling below 0.50, meaning the model is correct on at least half its flags while catching at least three quarters of real churners.

Model	Threshold	Recall	Precision	F1	AUC
Logistic Regression	0.30	0.757	0.523	0.619	0.84
Random Forest	0.25	0.807	0.513	0.627	0.82
XGBoost	**0.25**	**0.826**	**0.514**	**0.630**	**0.84**

4.5 Model selection

XGBoost at threshold 0.25 was selected as the final model based on:

- Highest recall (0.826) — catches 83% of real churners
- Highest F1 score (0.630) — best overall balance
- Highest average precision (0.66) on the precision-recall curve
- Tied highest AUC (0.84) with Logistic Regression
- Most informative feature importance output for business interpretation

Threshold tuning reduced missed churners from 199 to 65 (a 67% reduction) while the number of customers correctly identified as churners increased from 175 to 309.

4.6 Feature importance

XGBoost's feature importance scores identify which variables the model relied on most when making predictions. The top five drivers of churn are:

1. **Internet service_ fiber optic** (importance: 0.28): the single largest signal
2. **Contract type_two year** (importance: 0.20): strongly protective against churn
3. **Contract type_one year** (importance: 0.18): moderately protective
4. **Internet service_no internet** (importance: 0.07): no internet customers churn less
5. **Tenure** (importance: 0.04): longer tenure reduces churn probability

These findings are consistent across all three models, strengthening confidence in their validity. Three independent algorithms trained on the same data identified the same primary signals.

5. Retention Strategy & Recommendations

5.1 Tiered campaign structure

The model scores each customer with a churn probability between 0 and 1. We segment flagged customers into ***three tiers based on confidence level***, each with a distinct recommended action:

A. **High risk (probability ≥ 0.70) — 109 customers** These customers have a 70% or higher probability of churning. The model is highly confident. Delay is costly. Recommended action: immediate personal outreach, a phone call or direct message, with a concrete loyalty offer. Given that contract type is the primary driver, a discounted upgrade to a one-year contract is the highest-value intervention for this group.
B. **Medium risk (probability 0.50–0.70) — 213 customers** These customers are more likely than not to churn but the model is less certain. A retention call scheduled within two weeks is appropriate. Messaging should focus on

the value of their current services and explore any service quality concerns particularly for fiber optic customers in this tier.

C. **Low risk (probability 0.25–0.50) — 279 customers** These customers have an elevated probability relative to the general customer base but are not urgent. Include in the next scheduled email campaign with a soft retention message — a service highlight, a loyalty acknowledgement, or a feature reminder. No aggressive discount needed.

5.2 Recommended messaging by segment

Beyond the tier, the feature importance analysis suggests segment-specific messaging:

- ➔ **Fiber optic customers:** Lead with service quality acknowledgement. These customers are paying premium prices and likely have expectations around speed and reliability. A message that validates their investment and offers a service review or tech support check-in addresses the likely root cause rather than just offering a discount.
- ➔ **Month-to-month customers:** Lead with contract upgrade incentive. Offer a meaningful discount (e.g. two months free or 15% off) in exchange for signing a one-year contract. This converts the most at-risk billing structure into a more stable one and directly addresses the strongest churn driver.
- ➔ **Electronic check payers:** Offer a small incentive to switch to automatic payment, either a one-month discount or bill credit. This reduces payment friction and the monthly reconsideration moment that electronic check creates.
- ➔ **New customers (where tenure < 12 months):** Proactive welcome outreach at months 3 and 6. Not a retention call but a check-in. "How is your service?" before there is a problem to solve is more effective and less costly than damage control after dissatisfaction has formed.

6. Projected Business Impact

6.1 Campaign ROI by tier

The following projections are based on the assumptions stated in Section 6.3.

Tier	Customers	Campaign cost	Est. revenue saved	Net value	ROI
High risk	109	$1,635	$20,955	$19,320	1,082%
Medium risk	213	$3,195	$25,131	$21,936	806%
Low risk	279	$4,185	$25,953	$21,768	491%
Total	**601**	**$9,015**	**$72,039**	**$63,024**	**699%**

Every tier is ROI-positive. If budget is constrained, prioritising high-risk customers alone returns over $19,000 net on a $1,635 investment, that is a 1,082% return.

6.2 Sensitivity analysis

The 30% retention success rate is an industry benchmark. *Actual results may vary.* The table below shows net value at different retention rates:

Retention success rate	Est. revenue saved	Net value	ROI
20%	$48,026	$39,011	433%
25%	$60,033	$51,018	566%
30% (base case)	**$72,039**	**$63,024**	**699%**
35%	$84,046	$75,031	832%
40%	$96,052	$87,037	966%

Even at a conservative 20% retention success rate the campaign generates a 433% ROI. The business case is robust across a wide range of assumptions.

6.3 Assumptions

Assumption	Value	Basis
Average monthly revenue per customer	$64.76	Dataset mean of MonthlyCharges

Retention offer cost	$15.00 per customer	Standard telecom retention benchmark
Retention success rate	30%	Telecom industry benchmark
Customer lifetime horizon	12 months	Conservative estimate

These are estimates. Actual campaign performance should be tracked and used to recalibrate the model and assumptions on a quarterly basis.

7. Limitations & Next Steps

7.1 Limitations

1. **Dataset scope:** The analysis is based on a single US telecom provider dataset of 7,043 customers. Generalisation to other markets, geographies, or business models requires validation against local data. Results should be treated as directionally indicative rather than precisely predictive for a different customer base.

2. **Behavioural data gaps:** The dataset does not include customer support call logs, net promoter scores, app usage frequency, or complaint history. These variables are known to be strong churn predictors and their inclusion would likely improve model recall meaningfully.

3. **Assumed retention rate:** The 30% retention success rate is a benchmark, not a measured conversion rate for TelcoX specifically. The actual effectiveness of retention offers depends on offer design, delivery channel, timing, and customer service quality — none of which are captured in this analysis.

4. **Static model:** Customer behaviour changes over time. A model trained on historical data will gradually become less accurate as market conditions, competitive dynamics, and customer expectations shift. Regular retraining, quarterly at minimum, is recommended.

7.2 Next steps

➔ **Immediate:**

  - Deploy the retention campaign using retention_recommendations.csv as the target list.
  - Prioritise the 109 high-risk customers for personal outreach this week.

➔ **Short term (1–3 months):**
  - Track actual retention outcomes for contacted customers.
  - Measure the real conversion rate against the 30% assumption.
  - Use this data to recalibrate the cost-benefit projections.

➔ **Medium term (3–6 months):**
  - Enrich the model with additional data sources: support call frequency, app login data, NPS scores.
  - Retrain on the expanded feature set.
  - Explore whether a segment-specific model (e.g. a separate model for fiber optic customers) outperforms the general model.

➔ **Long term:**
  - Build a real-time scoring pipeline that flags newly at-risk customers as their behaviour changes rather than running the analysis periodically.
  - Integrate churn probability scores into the CRM so retention teams see risk scores alongside customer records.

Appendix A: Model Performance Detail

Baseline performance (threshold 0.50)

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.806	0.656	0.561	0.605	0.84
Random Forest	0.784	0.621	0.481	0.542	0.82
XGBoost	0.801	0.656	0.524	0.582	0.84

Tuned performance (optimal threshold)

Model	Threshold	Accuracy	Precision	Recall	F1
Logistic Regression	0.30	0.752	0.523	0.757	0.619
Random Forest	0.25	0.743	0.513	0.807	0.627
XGBoost	0.25	0.751	0.514	0.826	0.630

Confusion matrix for XGBoost at threshold 0.25

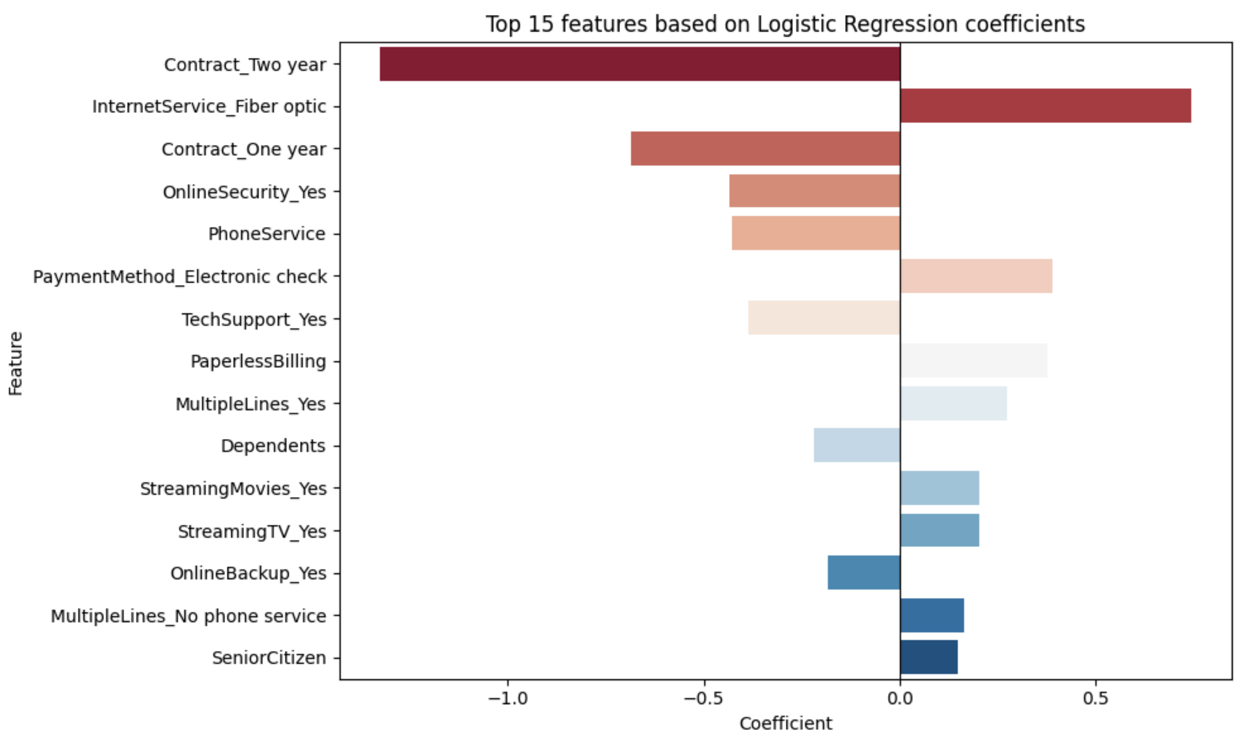
	Predicted: Stayed	Predicted: Churned

Actually stayed	743	292
Actually churned	65	309

Missed churners reduced from 199 (default threshold) to 65 (tuned threshold) — a 67% reduction.

Appendix B: Feature Importance Analysis

Logistic Regression coefficients

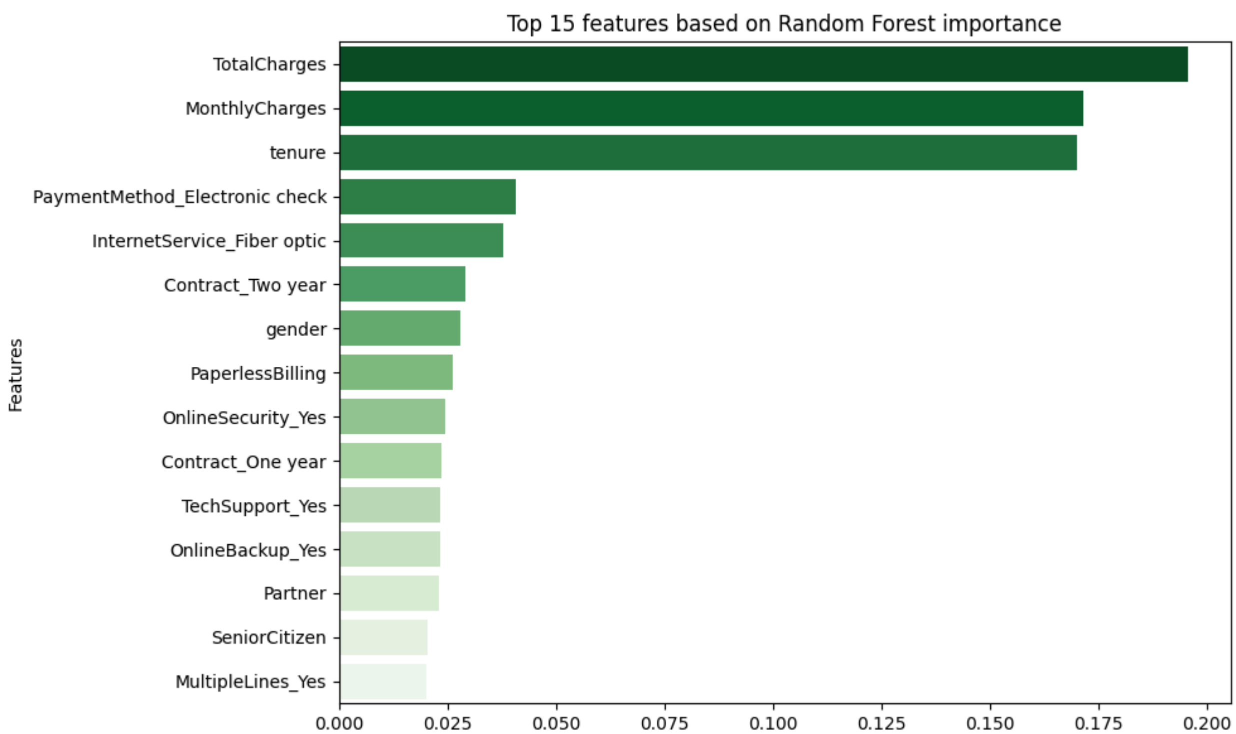


Logistic Regression coefficients indicate both the magnitude and direction of each feature's influence. Negative coefficients reduce churn probability; positive coefficients increase it.

Strongest protective factors (negative coefficients): *Contract_Two year (−1.45), Contract_One year (−0.72), OnlineSecurity_Yes (−0.48), TechSupport_Yes (−0.38)*

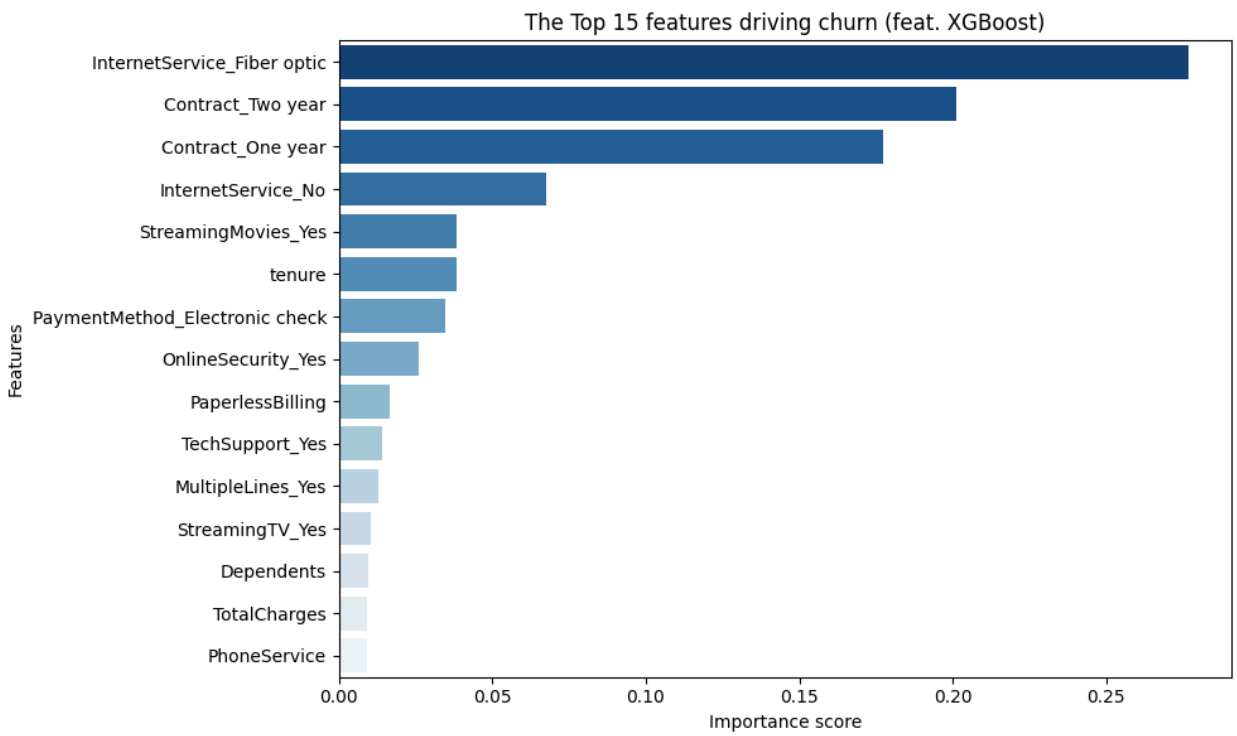
Strongest churn drivers (positive coefficients): *InternetService_Fiber optic (+0.84), PaymentMethod_Electronic check (+0.38), PaperlessBilling (+0.36), SeniorCitizen (+0.24)*

Random Forest feature importance



Random Forest heavily weighted three correlated financial variables — TotalCharges (0.196), MonthlyCharges (0.172), and tenure (0.168) — which together account for over 53% of total importance. This over-reliance on correlated variables likely explains Random Forest's weaker performance relative to the other two models.

XGBoost feature importance



XGBoost identified a more distributed and business-interpretable set of drivers, with contract type and internet service type dominating rather than raw financial variables. This distribution produces more actionable insights for retention strategy design.

Appendix C: Assumptions & Data Dictionary

Assumptions

Parameter	Value	Notes
Monthly revenue	$64.76	Mean of MonthlyCharges column
Offer cost	$15.00	Per customer contacted
Retention success rate	30%	Industry benchmark
Customer lifetime	12 months	Conservative
Test set size	1,409 customers	20% of full dataset
Flagged customers	601	At threshold 0.25

Key variable definitions

Variable	Description
Churn	Whether the customer left (1) or stayed (0)
Tenure	Number of months the customer has been with TelcoX
MonthlyCharges	Current monthly bill amount in USD
Contract	Month-to-month, one year, or two year

InternetService	DSL, fiber optic, or no internet service
PaymentMethod	Electronic check, mailed check, bank transfer, credit card
Churn_Probability	Model-assigned probability of churn (0–1)
Risk_Tier	High / Medium / Low based on probability thresholds
Est_Revenue_At_Risk	MonthlyCharges × 12 months per customer